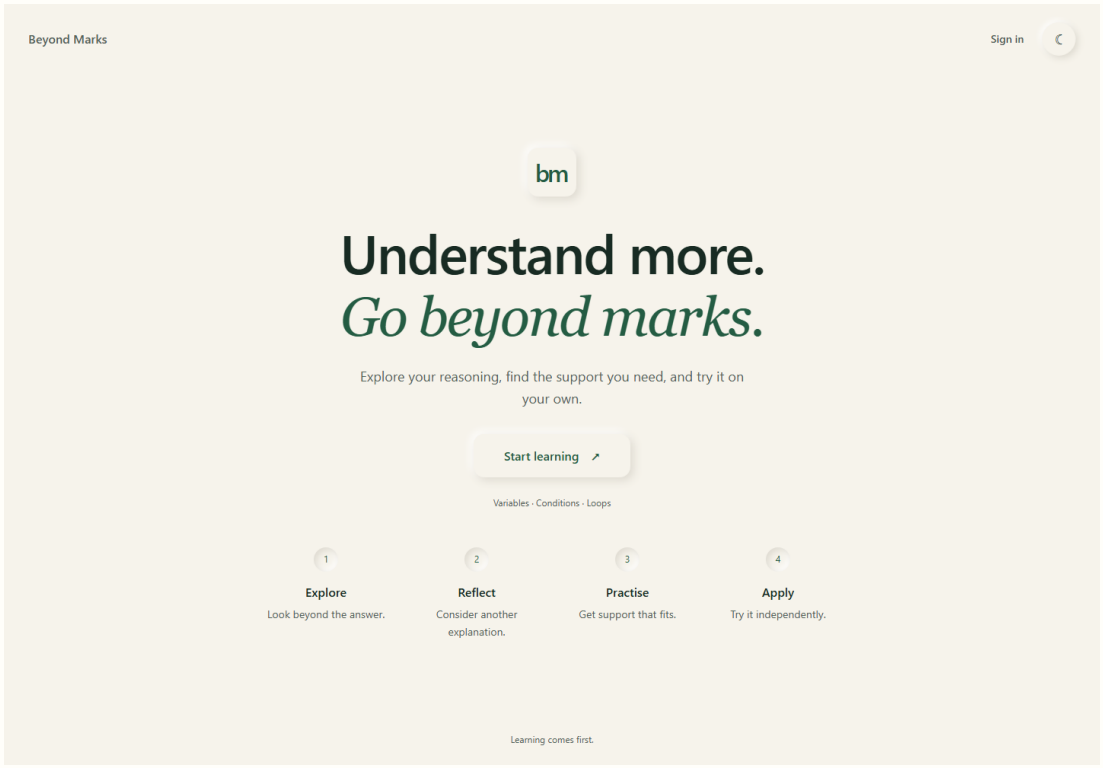
REASONING, AGENCY AND LEARNING OVER TIME

Understanding comes first

A persistent learning workspace for introductory programming learners and their teachers.



What the interface makes visible

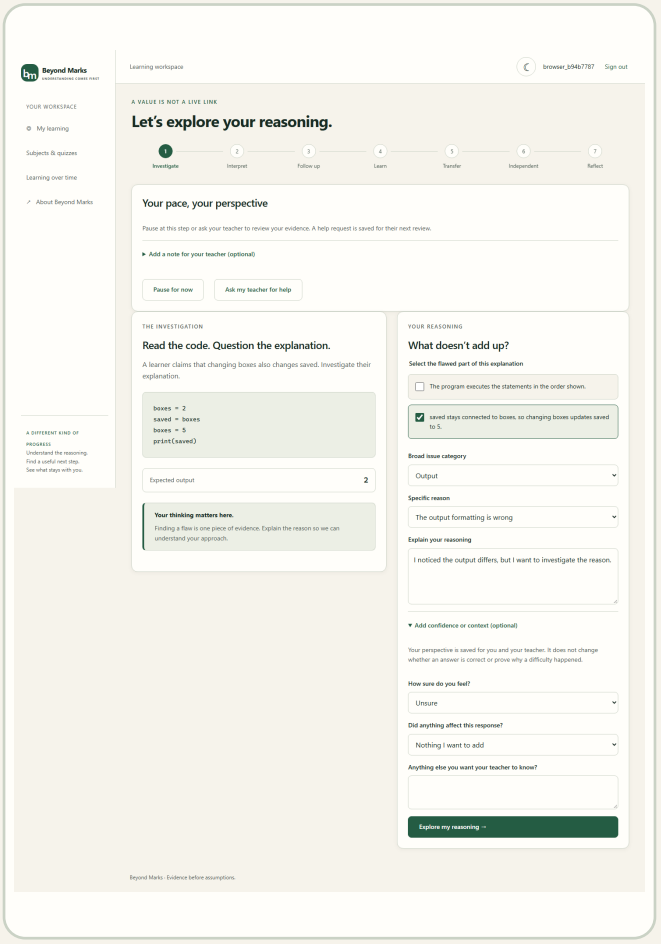
1. Reasoning behind an answer • 2. Competing interpretations • 3. A useful diagnostic probe • 4. Support, transfer and real delayed checks

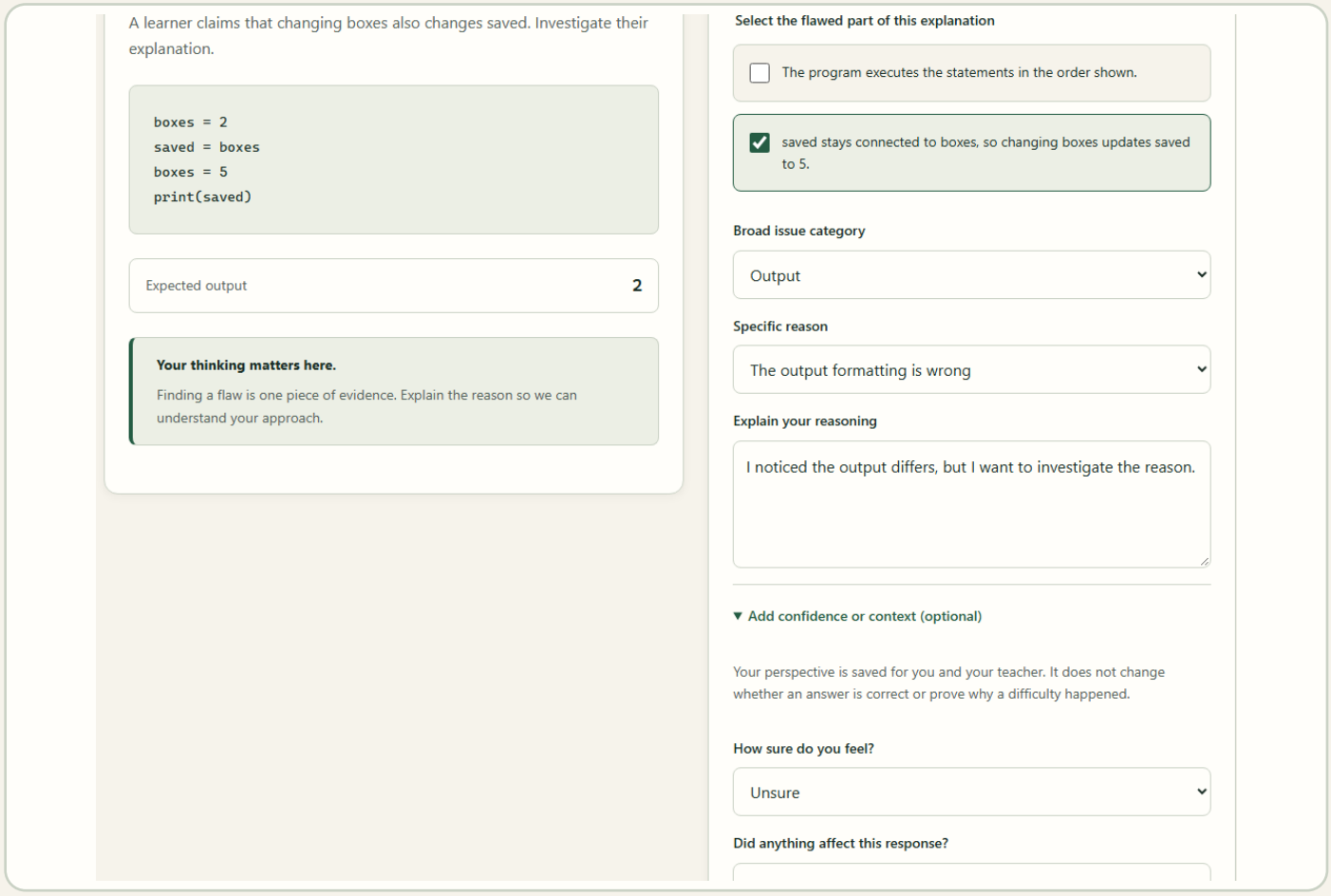
Teachers also publish class quizzes and append audited corrections to AI feedback. Learners can challenge an interpretation and request help.

This pack uses unchanged screenshots from the local Next.js/FastAPI/PostgreSQL application. The following pages pair full-screen context with enlarged excerpts. No capture is evidence of a genuine classroom pilot.

R1 · REASONING EVIDENCE

Investigate the explanation



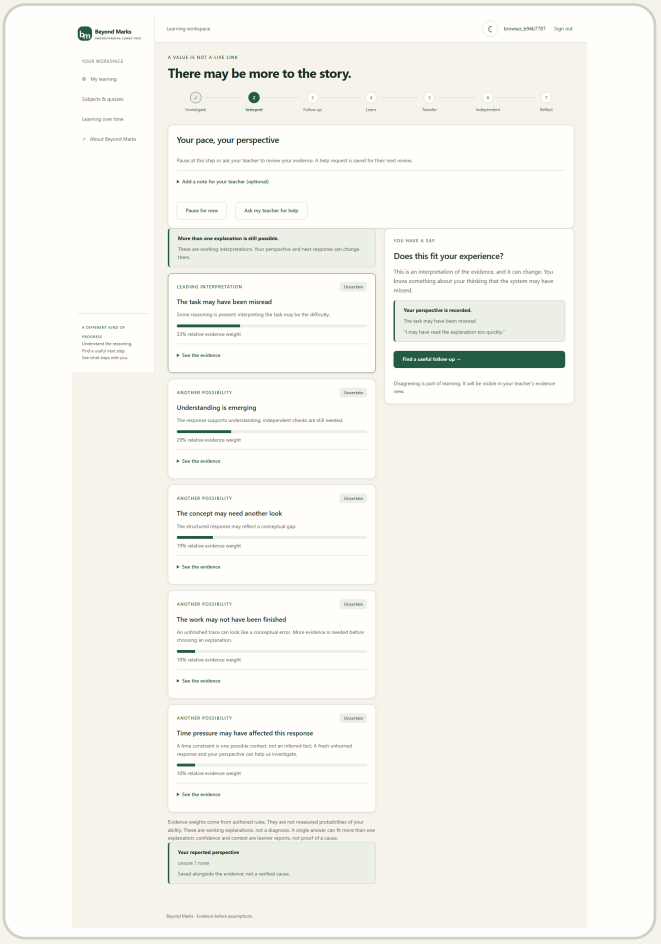


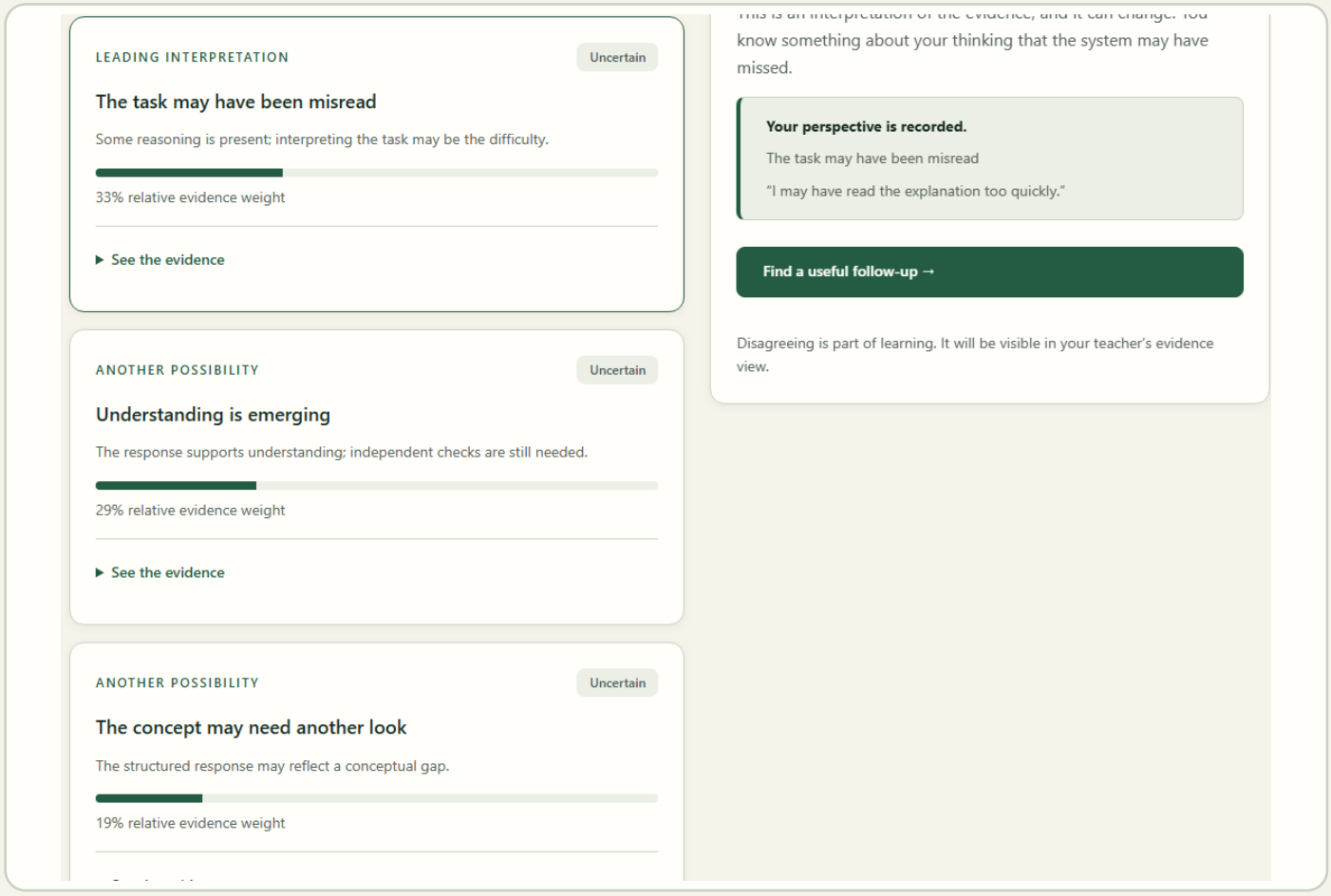
The learner identifies a flawed explanation, categorizes the issue and explains their thinking. Confidence and context add their perspective; a correct answer alone is not treated as proof of understanding.

Source: docs/screenshots/03-investigation-1440-light.png | Full screen at left; enlarged excerpt at right.

R2 · LEARNER AGENCY

Keep more than one explanation open



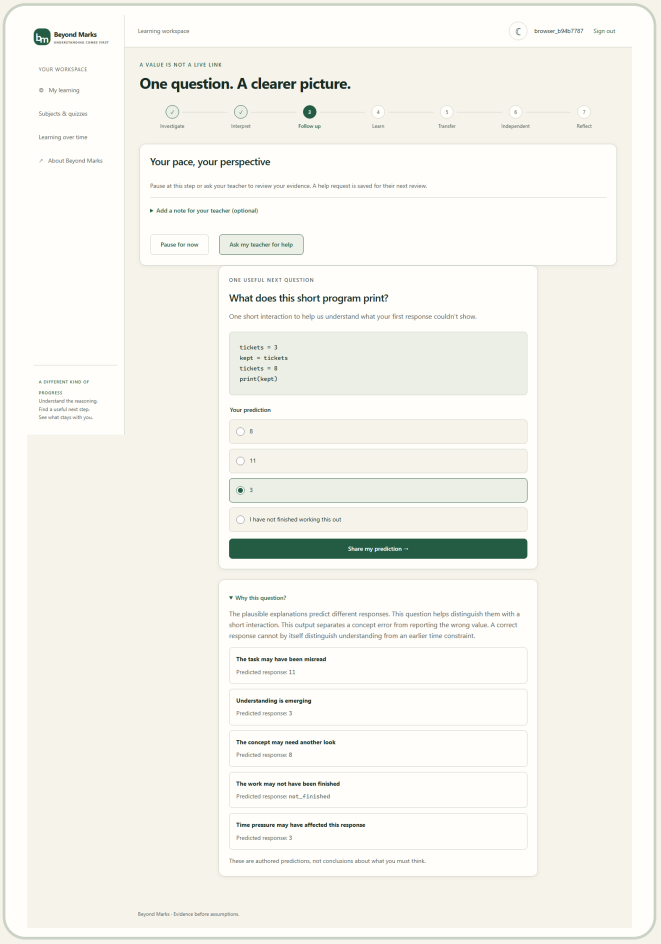


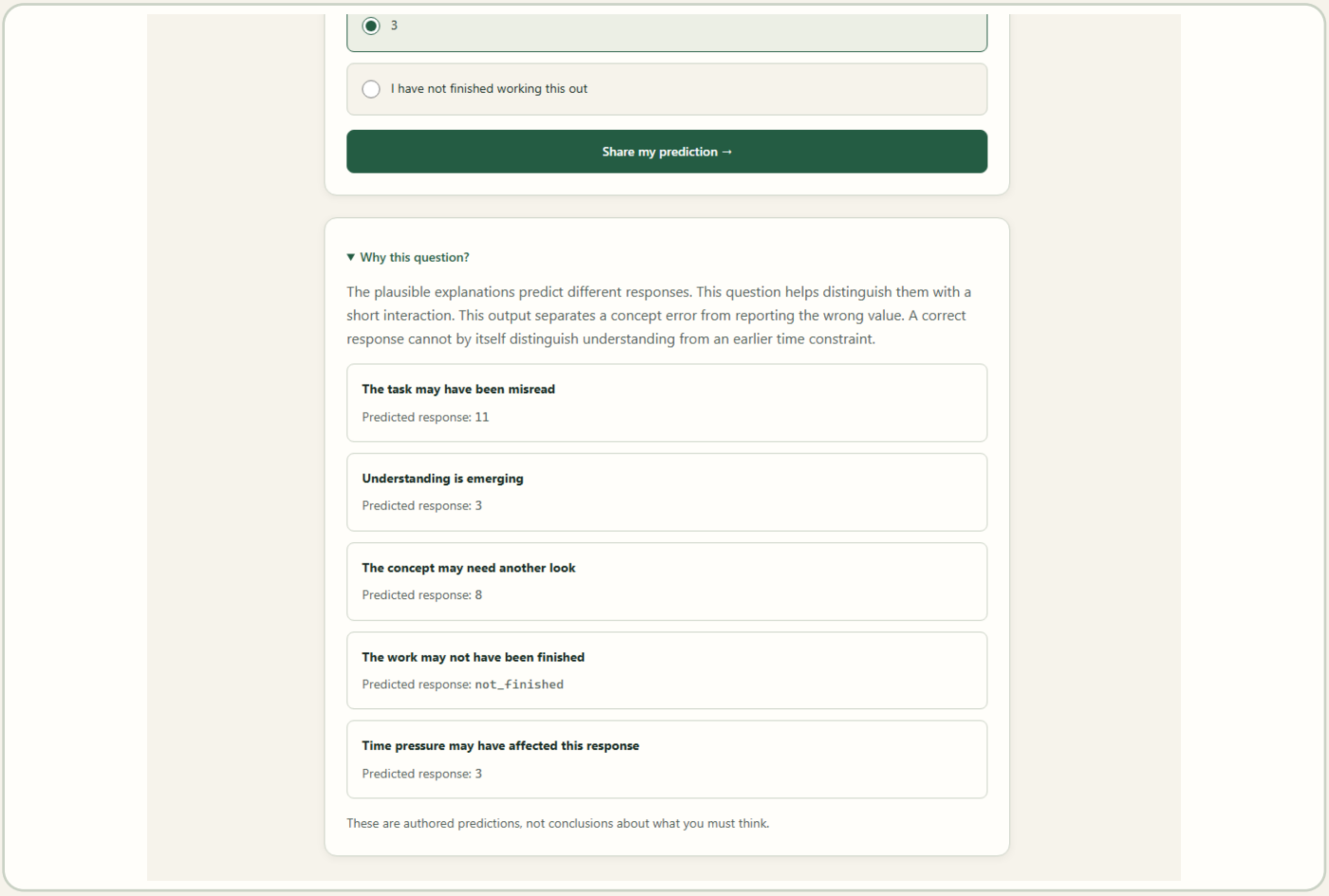
Competing interpretations remain provisional. Evidence summaries and learner disagreement preserve what the system believed and what the learner says. These authored weights are not probabilities of ability.

Source: docs/screenshots/04-interpretation-1440-light.png | Full screen at left; enlarged excerpt at right.

R3 · DIAGNOSTIC INTERACTION

Ask one useful next question



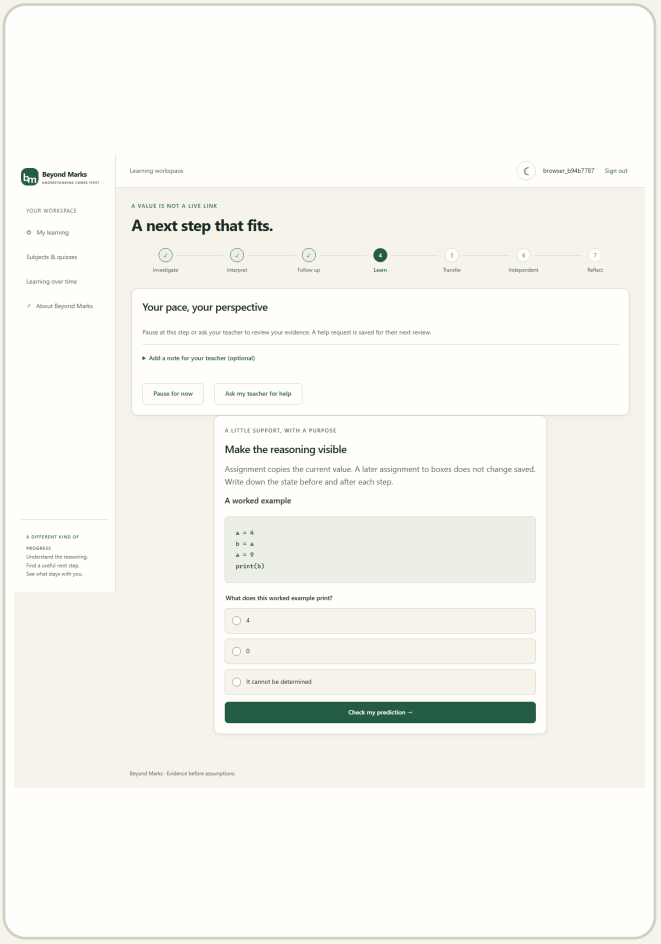


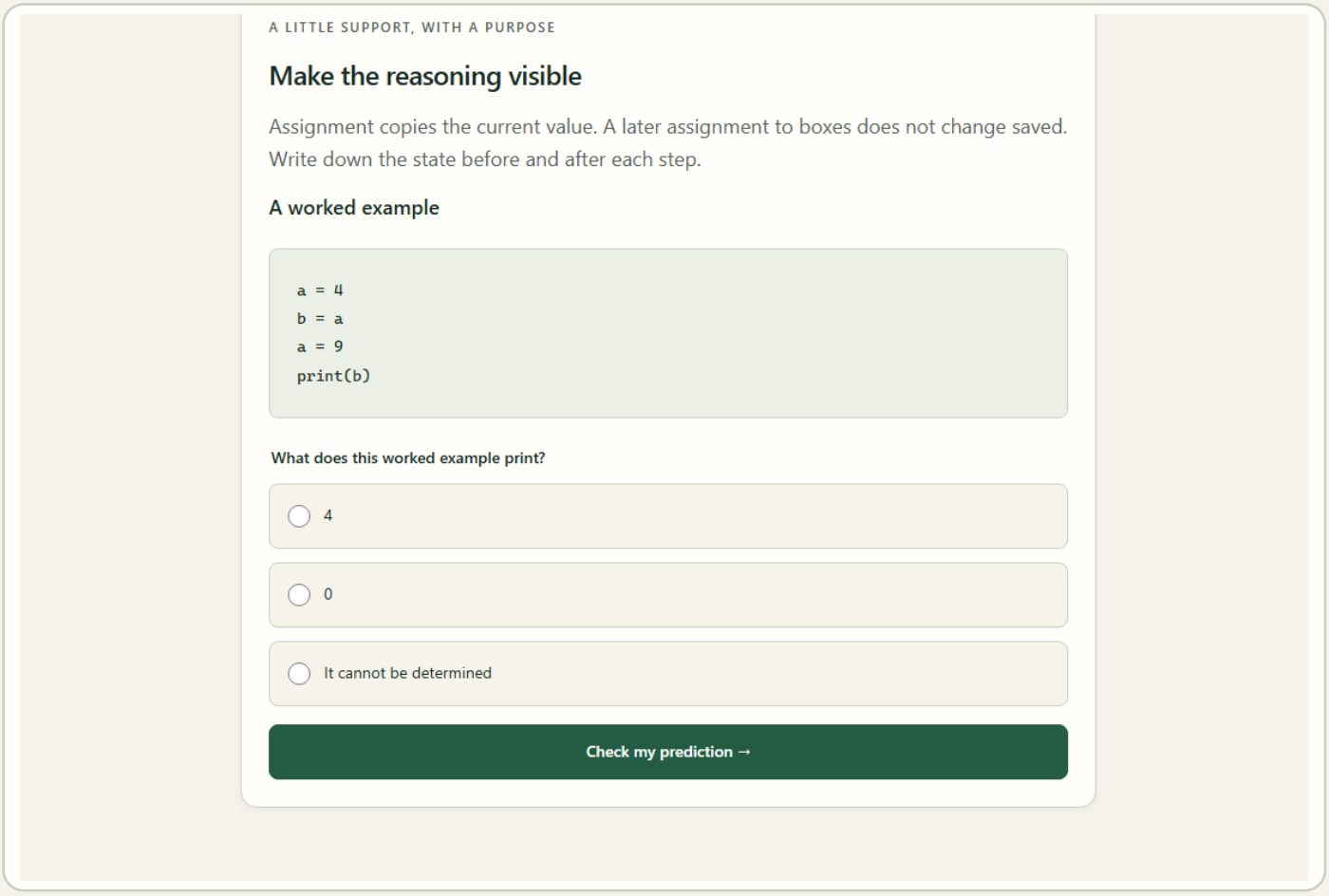
The selected probe asks for new evidence. Its rationale exposes competing predictions and the burden tradeoff. Different output, reading and unhurried probes test different possible explanations.

Source: docs/screenshots/05-follow-up-1440-light.png | Full screen at left; enlarged excerpt at right.

R4 · SUPPORTED LEARNING

Make the reasoning visible



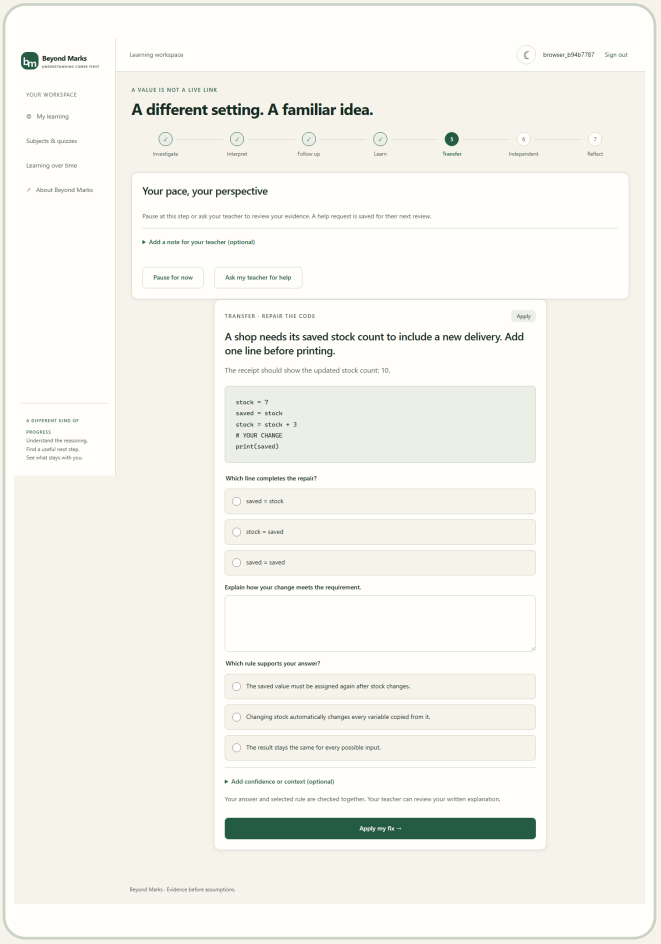


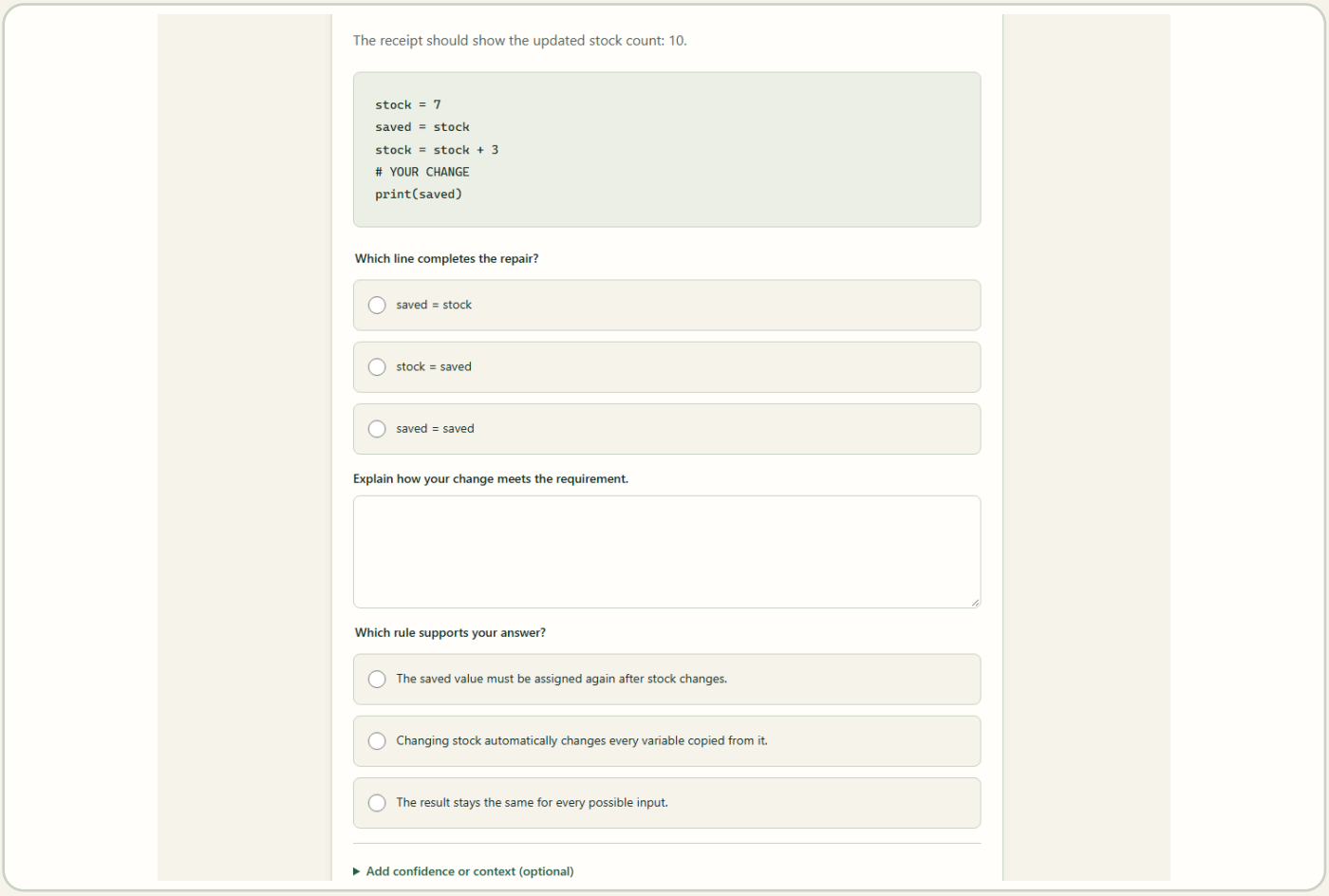
A worked example leads to a learner prediction and feedback. Learners read the feedback before continuing. A failed practice response leads to another example rather than an unsupported success label.

Source: docs/screenshots/11-support-1440-light.png | Full screen at left; enlarged excerpt at right.

R4 · TRANSFER

Apply the idea in a changed context



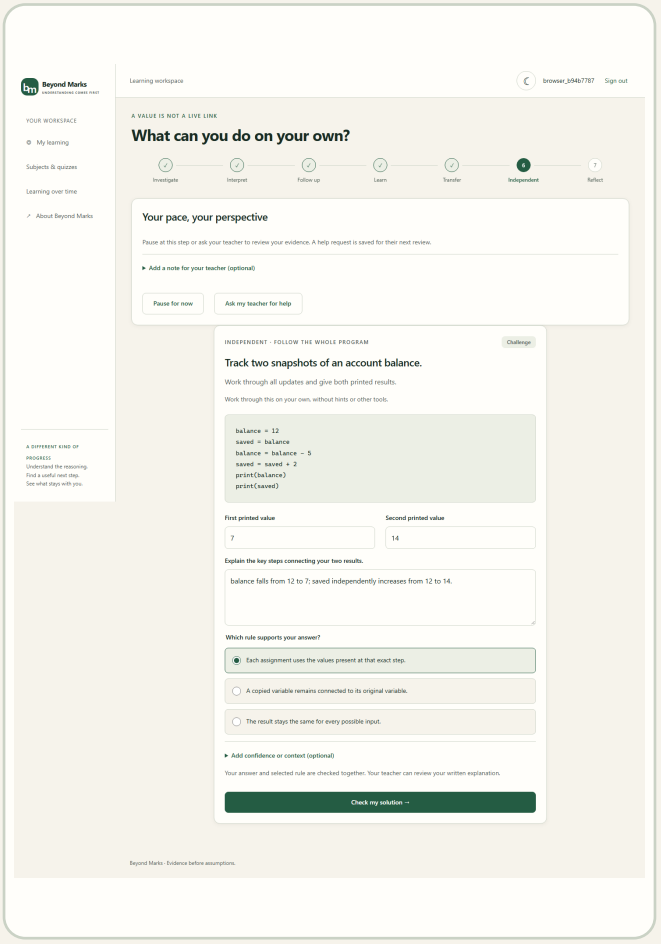


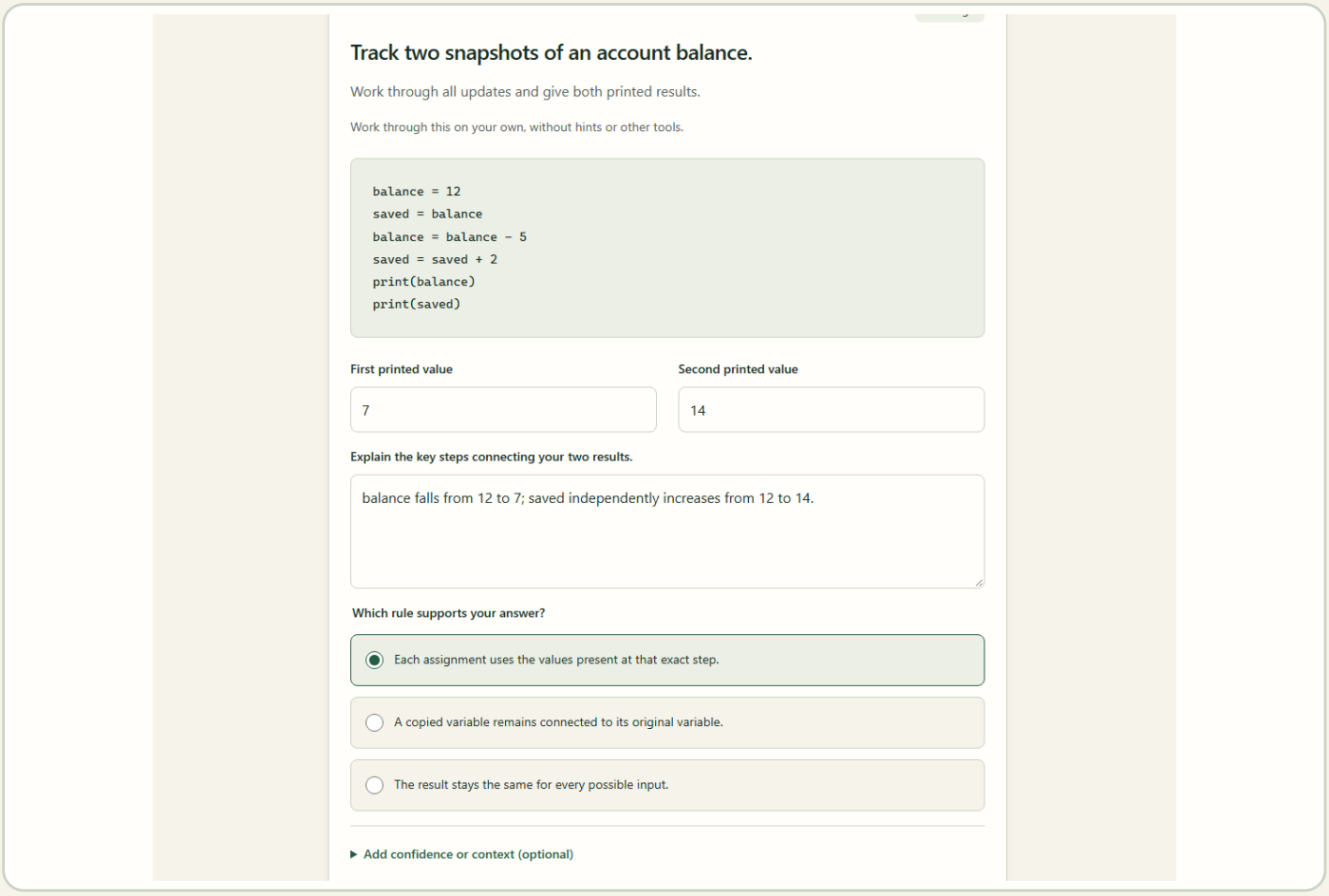
Transfer requires applying the rule to a different task. The learner supplies a repair, output and structured reasoning; failed checks return to support and fresh practice.

Source: docs/screenshots/12-transfer-1440-light.png | Full screen at left; enlarged excerpt at right.

R4 · INDEPENDENT CHECK

Check what the learner can do alone



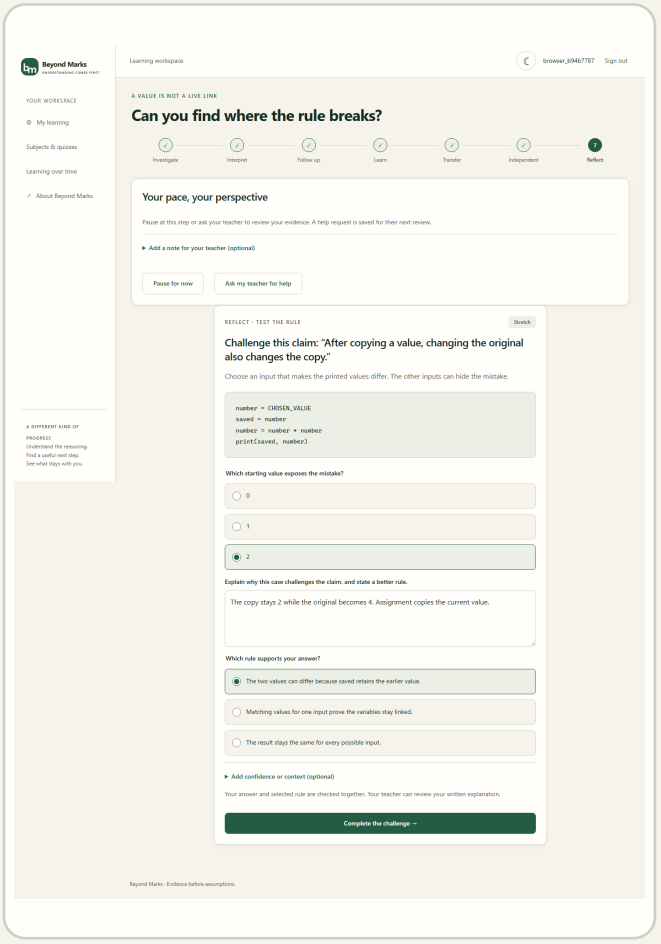


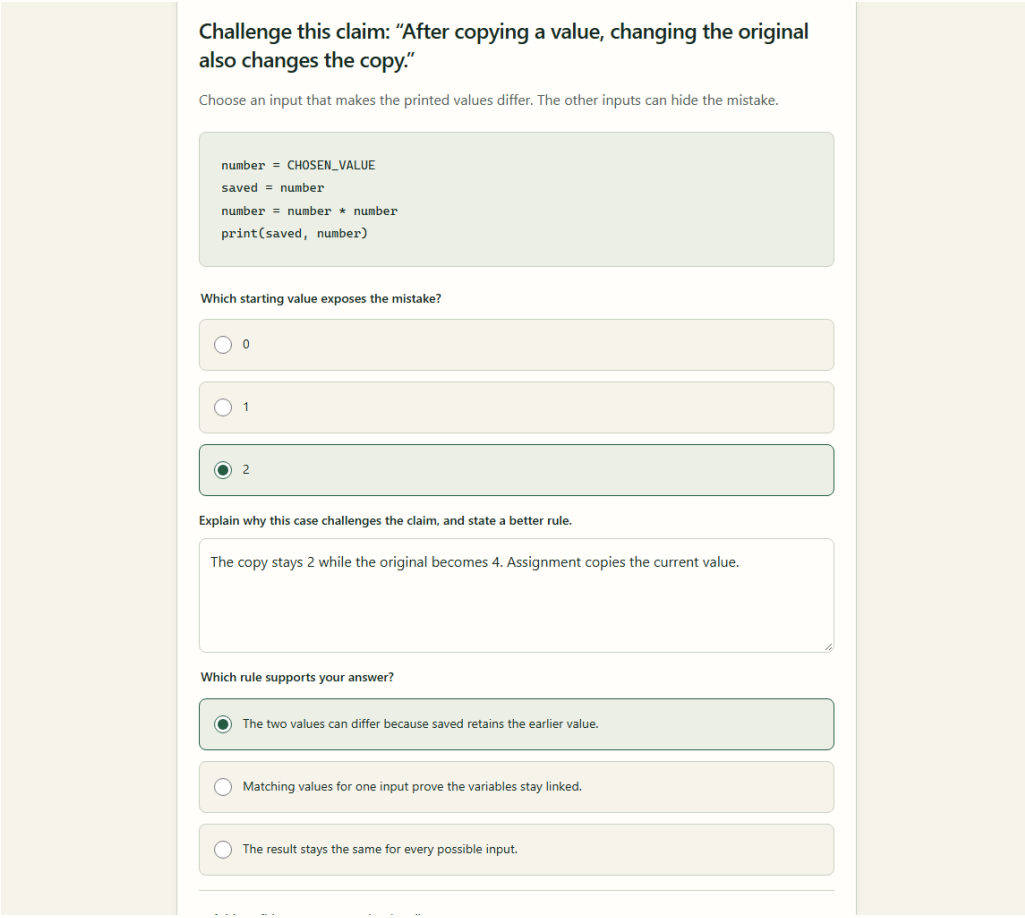
Independent work removes current hints and answer reveals. The learner supplies a solution and reasoning. Filled fields are demonstration responses, not measured participant outcomes.

Source: docs/screenshots/06-independent-1440-light.png | Full screen at left; enlarged excerpt at right.

R4 · REFLECTION

Challenge an overgeneralized claim



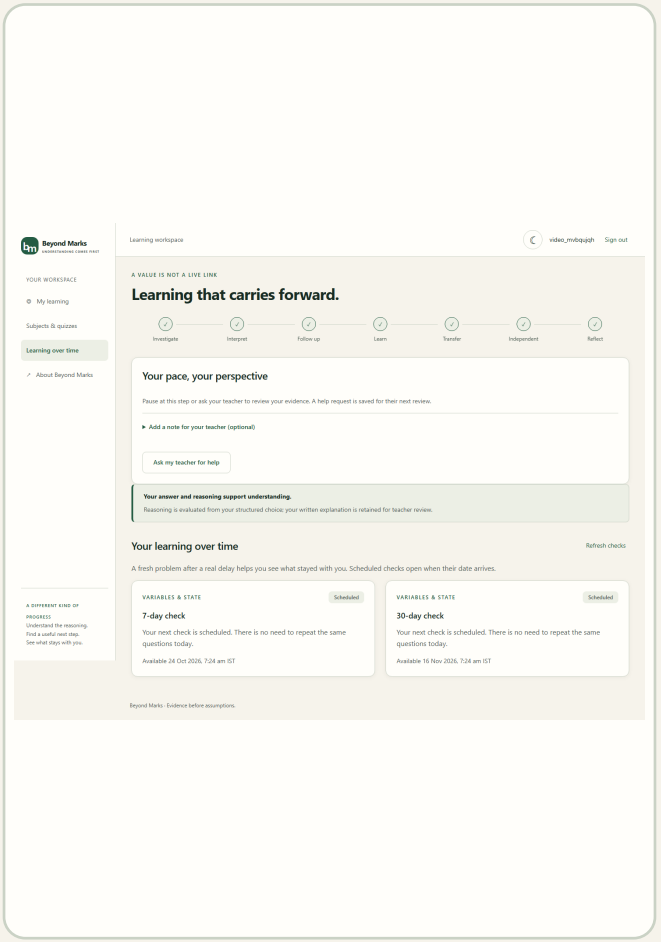


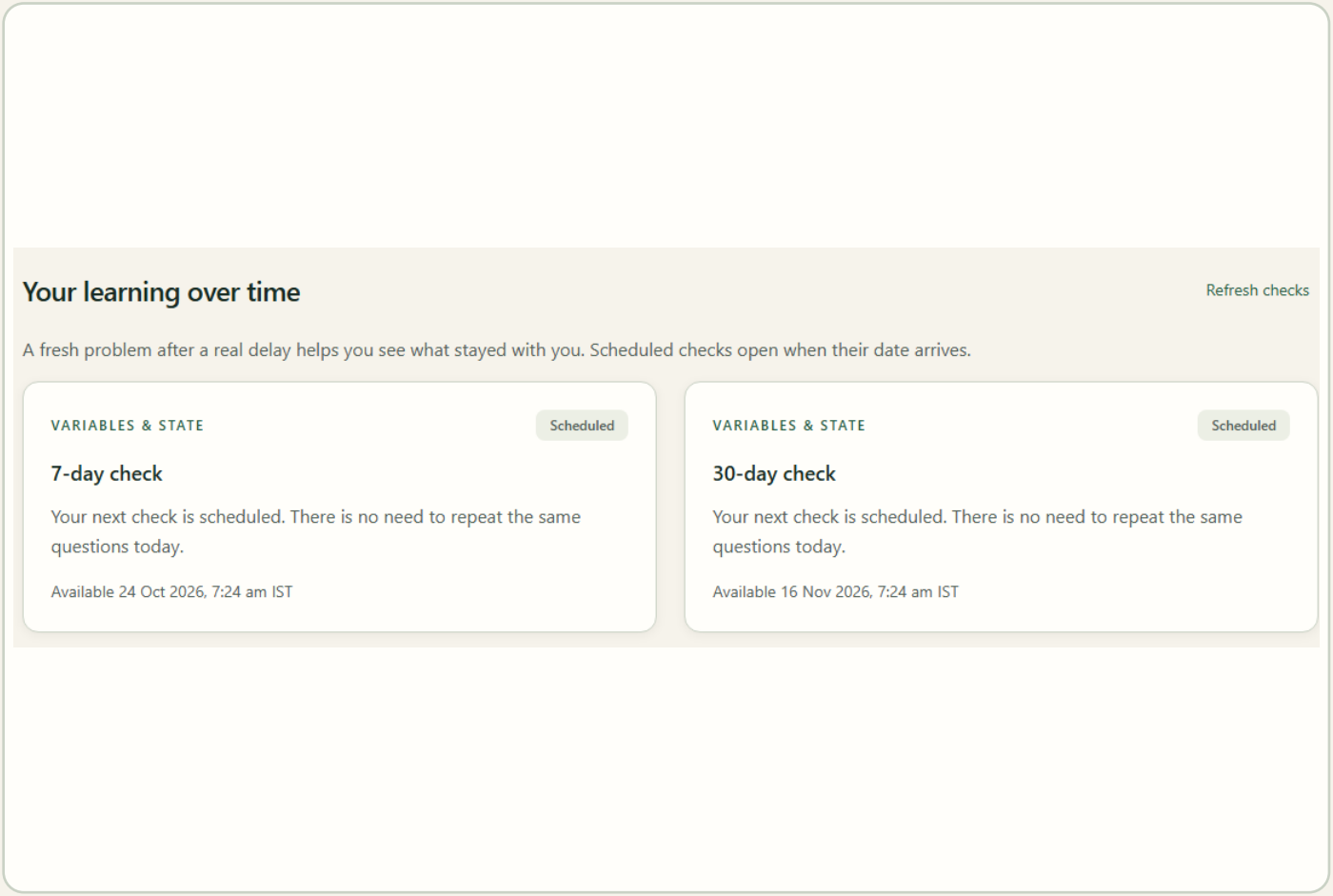
A reflection task asks the learner to test a claim and justify a counterexample. Completion requires structured evidence across the investigation, rather than advancing after an incorrect response.

Source: docs/screenshots/17-reflection-1440-light.png | Full screen at left; enlarged excerpt at right.

R4 · REAL ELAPSED TIME

Return after seven and thirty days



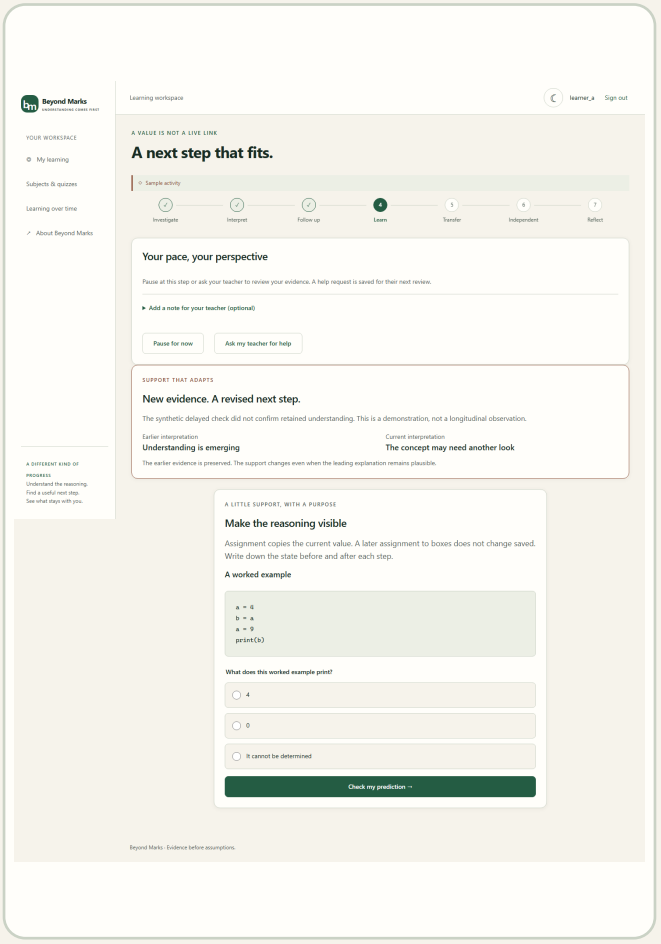


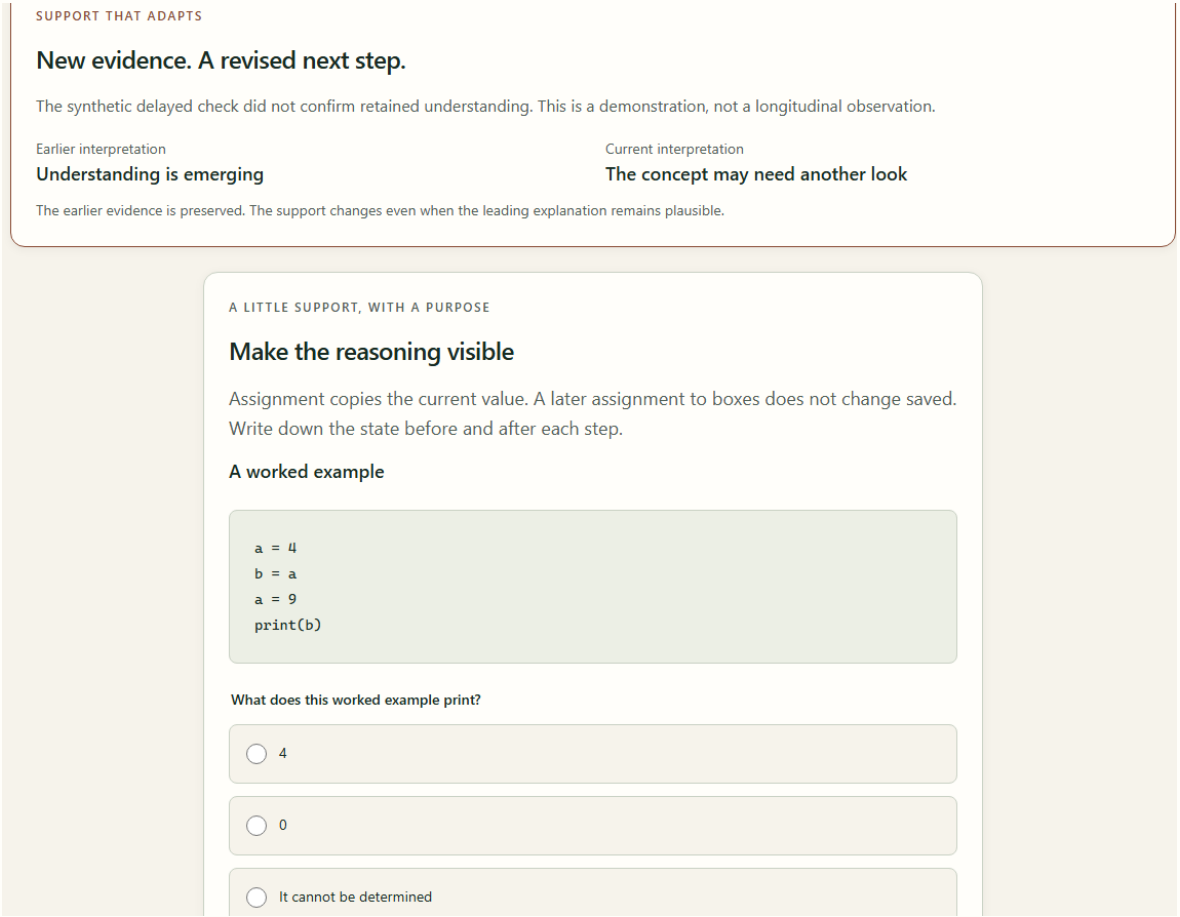
This actual recording account completed the flow on 17 October. The server scheduled checks for 24 October and 16 November 2026. These are future appointments, not completed retention outcomes. The account is demonstration activity, not a pilot participant.

Source: docs/screenshots/22-delayed-checks-1440-light.png | Full screen at left; enlarged excerpt at right.

R4 · ADAPTATION

Keep the earlier evidence when support changes



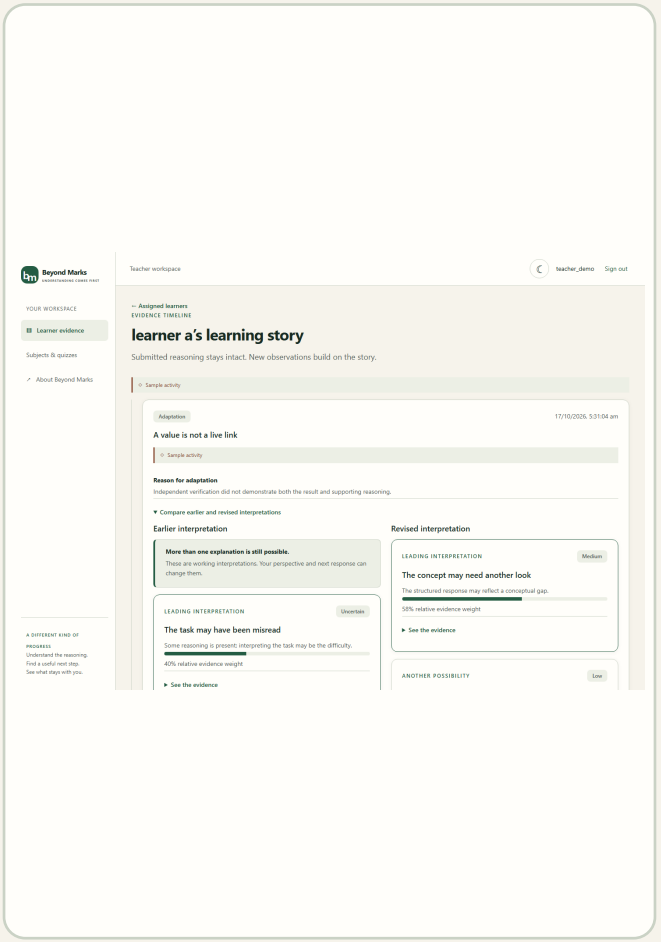


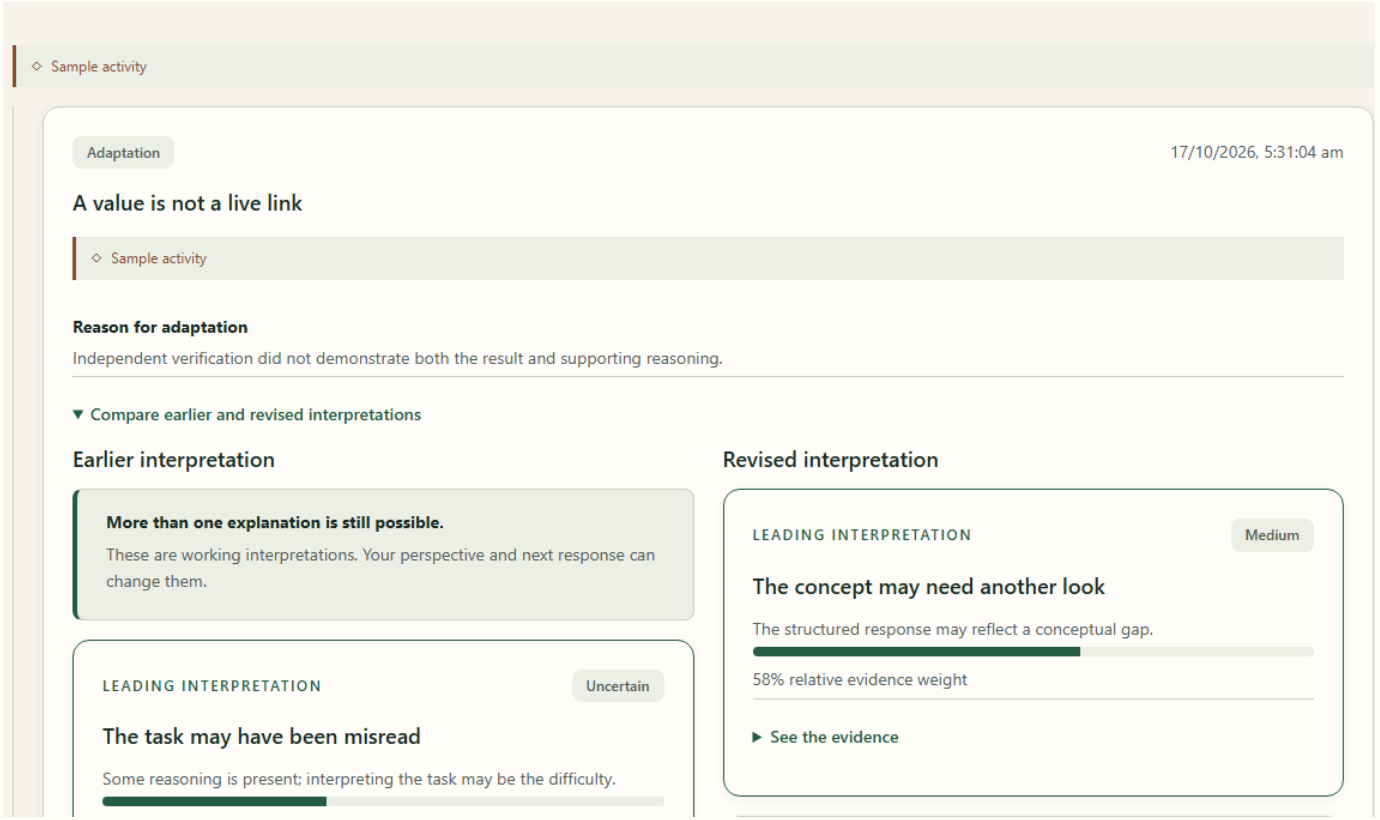
The adaptation view preserves earlier and revised interpretations and directs the learner to changed support. Historical seeded examples are synthetic demonstrations; genuine delayed failures use separately persisted observations.

Source: docs/screenshots/08-adaptation-1440-light.png | Full screen at left; enlarged excerpt at right.

TEACHER WORKSPACE

Review a learner’s reasoning history



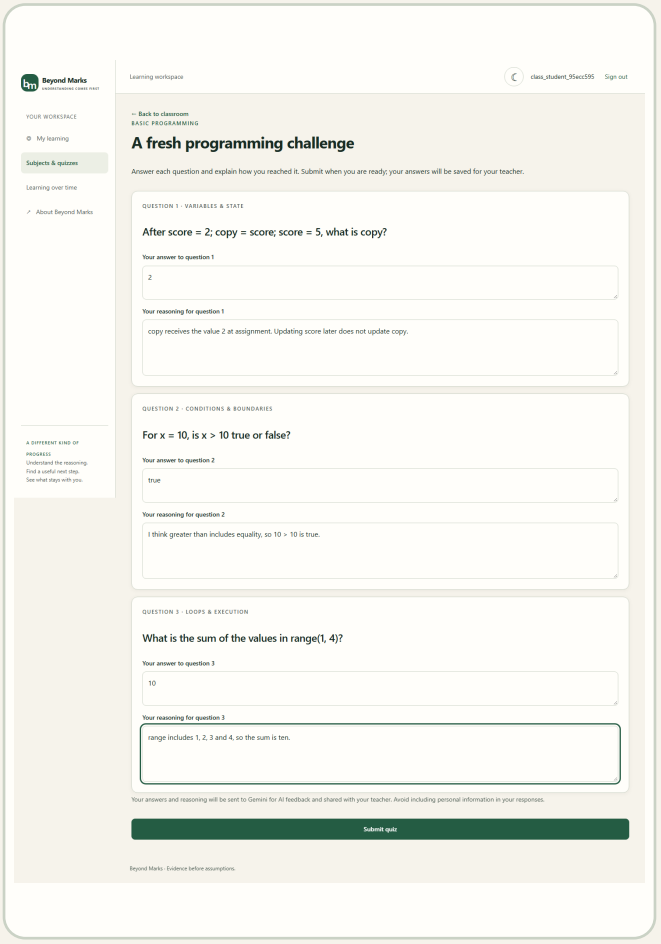


Only an assigned teacher can inspect the learner’s evidence. The timeline preserves earlier reasoning and interpretations; pagination loads bounded pages across evidence types.

Source: docs/screenshots/09-teacher-evidence-1440-light.png | Full screen at left; enlarged excerpt at right.

CLASSROOM EXTENSION

Create learning beyond the initial curriculum



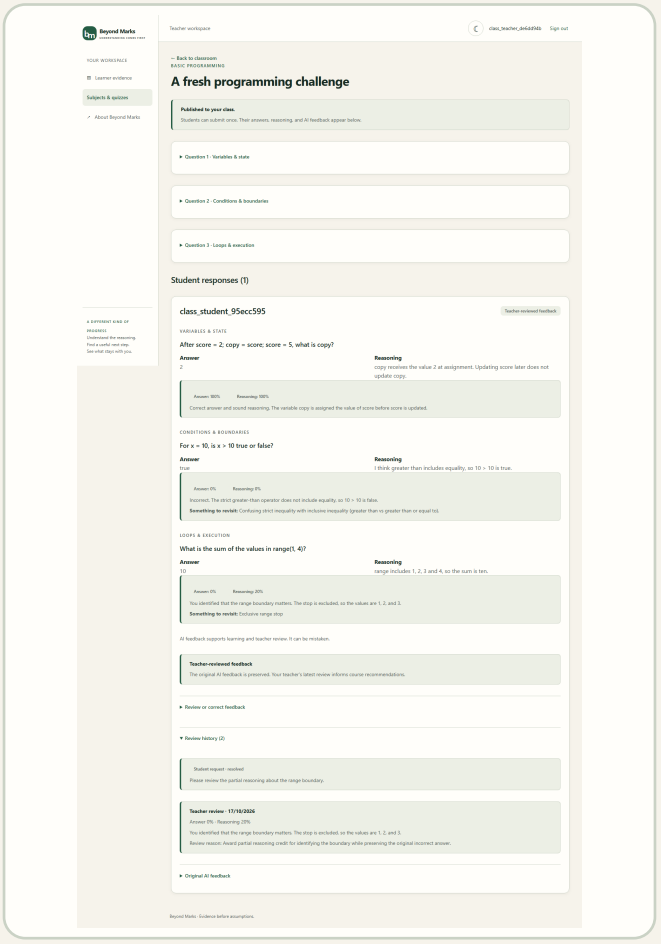


Teachers can add subjects and lessons, then publish quizzes with answer keys and reasoning guidance. Students see only published quizzes in their assigned class and submit answers with explanations.

Source: docs/screenshots/18-class-quiz-1440-light.png | Full screen at left; enlarged excerpt at right.

HUMAN REVIEW OF AI

Correct feedback without erasing its origin



AI feedback supports learning and teacher review. It can be mistaken.

Teacher-reviewed feedback

The original AI feedback is preserved. Your teacher's latest review informs course recommendations.

▶ Review or correct feedback

▼ Review history (2)

Student request · resolved

Please review the partial reasoning about the range boundary.

Teacher review · 17/10/2026

Answer 0% · Reasoning 20%

You identified that the range boundary matters. The stop is excluded, so the values are 1, 2, and 3.

Review reason: Award partial reasoning credit for identifying the boundary while preserving the original incorrect answer.

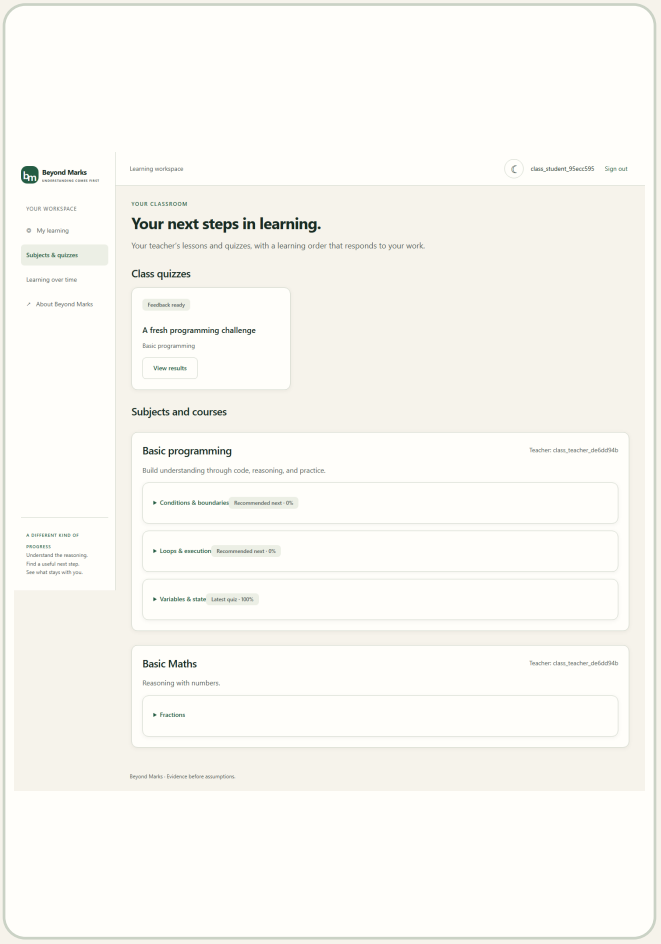
▶ Original AI feedback

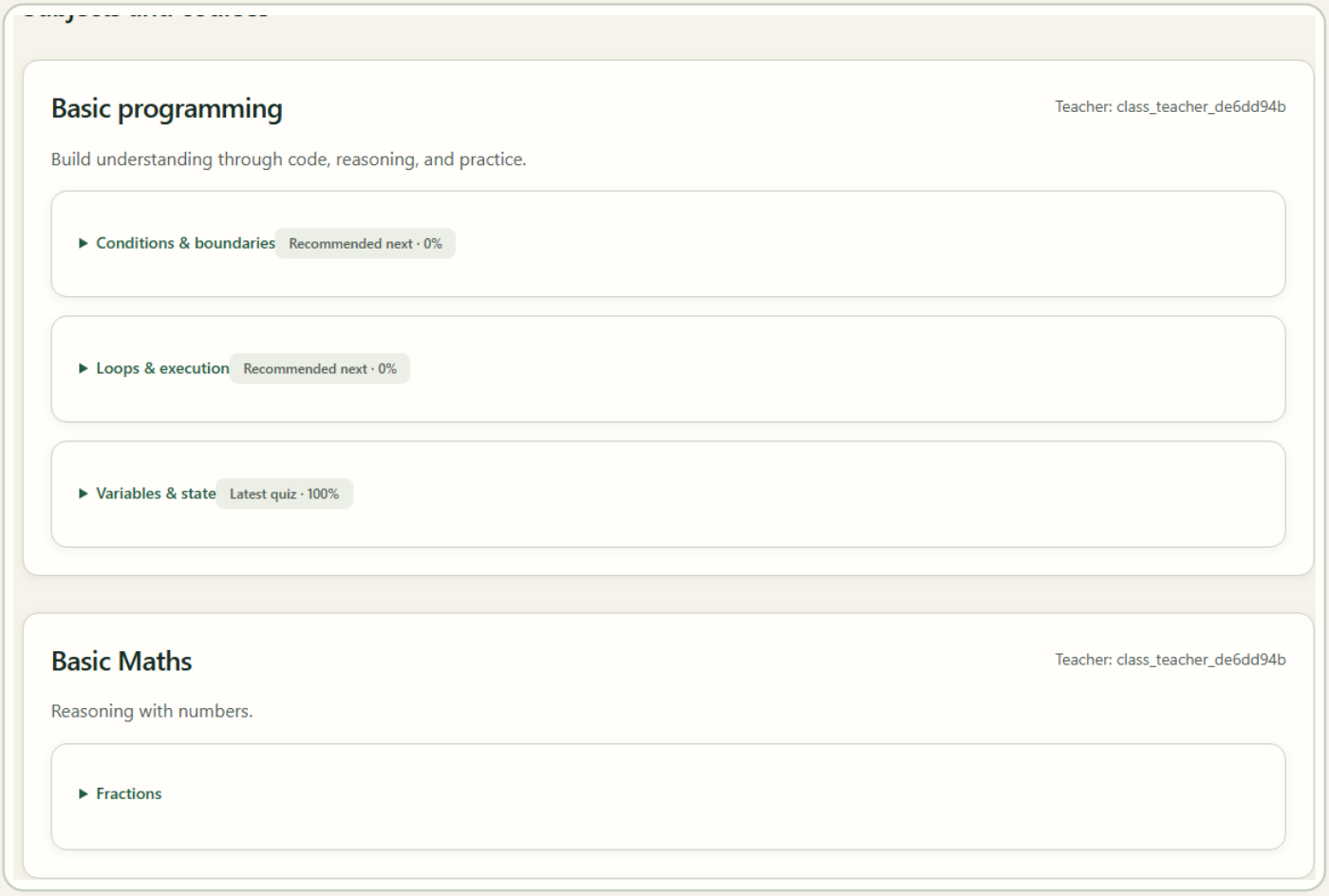
This actual teacher review awards partial reasoning credit while preserving an incorrect answer. A student concern is resolved, the teacher's reason stays in the audit trail, and the original AI feedback remains available.

Source: docs/screenshots/23-teacher-feedback-review-1440-light.png | Full screen at left; enlarged excerpt at right.

EVIDENCE-INFORMED NEXT STEPS

Recommend a course to revisit



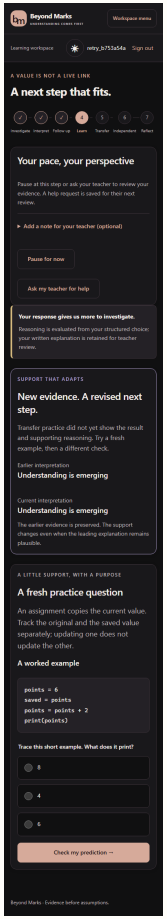


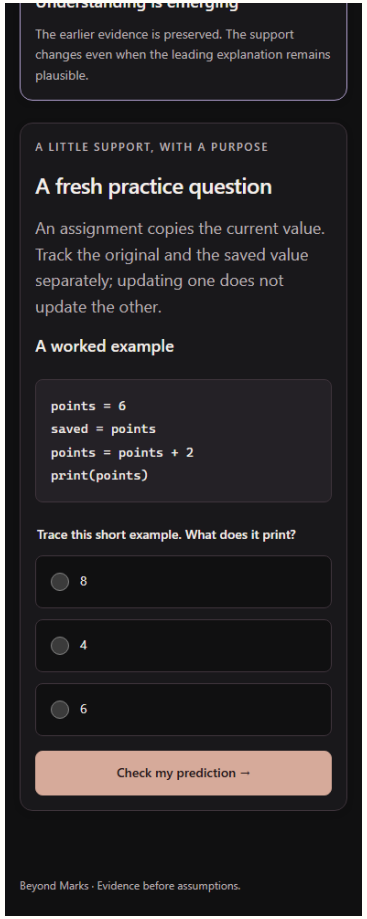
The classroom orders courses using the learner's latest eligible quiz evidence. Teacher corrections refresh the stored course summary. This is an individual learning recommendation, not a ranking of students.

Source: docs/screenshots/19-course-recommendations-1440-light.png | Full screen at left; enlarged excerpt at right.

RESPONSIVE INTERFACE

Keep practice usable on a phone





Full mobile screen and enlarged lower task section.

Dark theme: near-black surfaces, rose-gold primary actions and restrained accent colors. The design keeps task state and feedback visible.

The mobile dark theme retains the same saved task, feedback and recovery path. Screenshots are captured from the actual app at 390px and 1440px; presentation does not replace accessibility or classroom usability evaluation.

Source: docs/screenshots/21-fresh-practice-390-dark.png | Full screen at left; enlarged excerpt at right.

IMPLEMENTATION AND EVIDENCE

What this demonstration establishes

Implemented

Authored diagnosis, distinct probes, learner perspective and agency; interactive support with fresh practice; real 7/30-day scheduling and due-time enforcement; teacher-created lessons/quizzes; audited AI feedback corrections; schema-isolated pilot provisioning.

Evidence boundary

All shown accounts are demonstrations or automation fixtures. The scheduled dates show actual persisted future appointments. There are no fabricated elapsed-time outcomes and no completed longitudinal classroom pilot.

Next validation

Educator review of content and rules, consented classroom participation, actual delayed observations and measured false interpretations. Template difficulty, learning efficacy and production load remain unvalidated.

Review artifacts

Repository README: setup, upgrade/backfill instructions, risk limits and riddle mapping. Companion video: beyond-marks-demo.mp4. Public hosting has not been verified; the supported review path is local setup and downloadable artifacts.